

# Homework 7

**This homework assignment has six questions, continuing on a second page.**

Please read Section 1.5 (Expected Value) through the end of Example 1.22.

We ran out of time in class and did not discuss variance and standard deviation. We will talk about those concepts in our next class and practice with them on the next homework assignment.

1. The following counting problem has nothing to do with random variables! In Homework 5, you thought about splitting four people into two groups, where the order of the groups does not matter. There are three ways to do that. (See solutions for Homework 5 for details.) How many ways could you split six people into two groups of three where the order doesn't matter? How many ways could you split them into three groups of two where the order doesn't matter? Explain the principle of your counting method.

Splitting 6 people into two groups of 3:

Let's write  $N$  for the number of ways to split 6 people into two groups of 3 where the order does not matter. We know that there are  $6!/(3!3!) = 20$  ways to split 6 people into two groups of 3, if the order of the groups does matter. We can think of splitting into two groups in order as a two stage process: first split into two groups without an order ( $N$  ways) and then choose an order (permutation) of the two groups (2 ways). By the multiplication rule, we thus have  $N(2) = 20$ , and so  $N = 10$ .

Splitting 6 people into three groups of 2:

Let's write  $M$  for the number of ways to split 6 people into three groups of 2 where the order does not matter. We know that there are  $6!/(2!2!2!) = 90$  ways to split 6 people into three groups of 2, if the order of the groups does matter. We can think of splitting into three groups in order as a two stage process: first split into three groups without an order ( $M$  ways) and then choose an order (permutation) of the three groups ( $3!$  ways). By the multiplication rule, we thus have  $M(3!) = 90$ , and so  $M = 15$ .

2. Consider the chance experiment of rolling a red die and a green die. Let  $X$  be the random variable that assigns to each outcome the higher of the two rolls. For example,  $X(5, 3) = 5$ ,  $X(2, 2) = 2$ , and  $X(1, 6) = 6$ . You have calculated the distribution of this random variable. The possible values of  $X$  are 1, 2, 3, 4, 5, 6, and  $X$  takes on these values with the following probabilities:  $P(X = 1) = 1/36$ ,  $P(X = 2) = 3/36$ ,  $P(X = 3) = 5/36$ ,  $P(X = 4) = 7/36$ ,  $P(X = 5) = 9/36$ ,  $P(X = 6) = 11/36$ .

Calculate  $E[X]$ , the expected value of the higher of the two rolls.

We use the formula for  $E[X]$ :

$$E[X] = (1)(1/36) + 2(3/36) + 3(5/36) + 4(7/36) + 5(9/36) + 6(11/36) = 161/36 \approx 4.5.$$

If you repeat this chance experiment many times, you can expect that on average your higher roll is about 4.5 ( $161/36$ ).

3. Consider the chance experiment of flipping a coin five times. Let  $N$  be the random variable that assigns to each outcome the number of heads. For example,  $N(\text{HTHHT}) = 3$ . The possible values of  $N$  are 0, 1, 2, 3, 4, 5, and  $N$  takes on these values with the following probabilities:  $P(N = 0) = 1/32$ ,  $P(N = 1) = 5/32$ ,  $P(N = 2) = 10/32$ ,  $P(N = 3) = 10/32$ ,  $P(N = 4) = 5/32$ ,  $P(N = 5) = 1/32$ .

Calculate  $E[N]$ , the expected value of the number of H's.

We use the formula for  $E[N]$ :

$$E[N] = (0)(1/32) + (1)(5/32) + (2)(10/32) + (3)(10/32) + (4)(5/32) + (5)(1/32) = 80/32 \text{ or } 2.5.$$

If you repeat this chance experiment many times, you can expect that on average you get 2.5 heads.

After reading about binomial distributed random variables in Chapter 2, we can solve this problem in a different way. We have  $N \sim \text{binomial}(5, 1/2)$  (i.e.  $N$  has the binomial(5, 1/2) distribution). According to the formulas in Chapter 2, if a random variable has the binomial( $n$ ,  $p$ ) distribution, its expected value is  $np$ . In this case, we may conclude that  $E[N]$  is  $(5)(1/2) = 2.5$ .

4. Please work Exercise 40 from Durrett.

Let  $X$  be the random variable that is the person's payoff playing the game. The random variable  $X$  has two values:

- a number  $x$  (the payout to be determined) if the person rolls a total of 7 on the two dice (probability  $1/6$ , as we have calculated earlier);
- $-1$  if the person does not roll a total of 7 on the two dice (probability  $5/6$ ).

$$E[X] = x(1/6) + (-1)(5/6).$$

To make the game fair, we need  $E[X] = 0$ :

$$0 = x(1/6) + (-1)(5/6)$$

$$5 = x.$$

You should pay the person \$5 to make the game fair.

5. Please work Exercise 41 from Durrett. You can interpret his description of the game as follows: Let  $X$  be the random variable telling your winnings or losses when you play this game. The possible values for  $X$  are 3 (win \$3) and  $-1$  (lose \$1). If  $p$  is the probability of winning,  $P(X = 3)$  is  $p$ , and  $P(X = -1)$  is  $1 - p$ . Durrett is asking you to find the value of  $p$  that makes  $E[X] = 0$  (fair game). Durrett writes in the back of the book that  $p$  is  $1/4$ . Explain how to figure this out.

Following my suggestions above, we calculate  $E[X]$  in terms of the unknown probability  $p$ :

$$E[X] = 3p + (-1)(1 - p) = 4p - 1.$$

For the game to be fair, we must have  $E[X] = 0$  and thus (by algebra)  $p = 1/4$ .

6. Please work Exercise 42 from Durrett.

You buy a ticket for the lottery. There are 100 tickets sold. The people running the lottery pick eight tickets. The first pick gets \$100, the second two get \$25 each, and the other five get \$10.

Let  $x$  be the amount you pay for a ticket, and let  $X$  be the random variable indicating how much you win in the lottery. The possible values for  $X$  are as follows:

$100 - x$  with probability  $1/100$

$25 - x$  with probability  $2/100$

$10 - x$  with probability  $5/100$

$-x$  with probability  $92/100$

We thus have  $E[X] = (100 - x)(1/100) + (25 - x)(2/100) + (10 - x)(5/100) + (-x)(92/100)$ . Simplifying that expression, we find  $E[X] = 1 + 50/100 + 50/100 - x = 2 - x$ . The lottery is a fair bet if  $E[X] = 0$ . That happens when  $x = 2$ . You should be willing to pay at most \$2 for a ticket (to make it a fair game or to your advantage).

There are other ways to reason out the fair price of \$2. This method has the advantage of matching the general reasoning about fair bets (random variables with expected value 0) that we used in the other two exercises from Durrett.